

Name: BRANDON MAROLT

Math 3108 – Linear Algebra
Exam 1
February 10, 2017

Directions:

- 1) All cell phones and other electronic noisemaking devices must be turned off or completely silenced (i.e. not on vibrate) for the duration of the exam.
- 2) No calculators, books, or other materials are permitted.
- 3) Show **ALL** your work! Correct answers which are not properly justified will not receive full credit.
- 4) Failure to follow directions specific to a problem will result in the loss of points.
- 5) Circle, box, or underline each final answer if necessary for clarity. Label all graphs.
- 6) If you work a problem two different ways, clearly indicate which one you want us to grade, preferably by crossing out the one you do not want us to look at. If you use two methods, one of which is wrong, and neither method is crossed out, you will not receive full credit.
- 7) Answers must be exact (like $\sqrt{2}$), not approximate (like 1.414), unless a problem specifically indicates otherwise.
- 8) Simplify where appropriate. Quantities such as $\sqrt{9}$ and $\cos \pi$ should be calculated.
- 9) If you need extra room, you may use the back of the previous page. However, you must indicate you are doing so by clearly writing "BPP" on the relevant problem.
- 10) This packet has six sheets of paper, including this cover page. Do NOT remove the staple or remove any sheet from this packet.
- 11) Once this exam begins, you will have 50 minutes to complete your solutions.

Page	1	2	3	4	5	Sum
Points Earned	12	14	18	15	11	70
Maximum Points	18	20	18	22	22	100

**DO NOT OPEN THIS EXAM
UNTIL TOLD TO DO SO**

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Multiple Choice and True/False. Circle the correct answer.

1. (8 points) Consider the matrix $A = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 4 \end{bmatrix}$.

a) A is in _____.

(circle one)

Echelon Form

Reduced Echelon Form

Neither

b) The equation $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution.

(circle one)

True

False

b) The transformation $T(\mathbf{x}) = A\mathbf{x}$ is _____ and _____.

(circle one)

One-to-One

Not One-to-One

(circle one)

Onto

Not Onto

2. (6 points) Consider the matrix $B = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 5 & 3 \\ 0 & 0 & -4 \\ 0 & 0 & 4 \end{bmatrix}$.

a) B is in _____.

(circle one)

Echelon Form

Reduced Echelon Form

Neither

b) The transformation $T(\mathbf{x}) = B\mathbf{x}$ is _____ and _____.

(circle one)

One-to-One

Not One-to-One

(circle one)

Onto

Not Onto

3. (2 points) The equation $A\mathbf{x} = \mathbf{b}$ is guaranteed to be consistent if the _____ has a pivot position in every row.

a)

augmented matrix $[A : \mathbf{b}]$

b)

coefficient matrix A

c)

neither of these

4. (2 points) If $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^n$, then the vector $-\mathbf{u}$ is in $\text{Span}\{\mathbf{u}, \mathbf{v}\}$.

a)

Always True

b)

Sometimes True

c)

Never True

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Open Answer. Show ALL your work!

5. Consider the system of linear equations

$$x_1 + 3x_2 = 2$$

$$4x_1 + hx_2 = 8$$

a) (4 points) Find the value(s) of h , if any, such that the system has exactly one solution.

$$\left[\begin{array}{cc|c} 1 & 3 & 2 \\ 4 & h & 8 \end{array} \right] \xrightarrow{4R_1 - R_2 \rightarrow R_2} \left[\begin{array}{cc|c} 1 & 3 & 2 \\ 0 & 12-h & 0 \end{array} \right]$$

$h \neq 12$

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b) (4 points) Find the value(s) of h , if any, such that the system has infinitely many solutions.

~~no values~~ for h exist

0

6. (8 points) Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation. If T is the vertical shear transformation that maps $\mathbf{e}_1$ into $\mathbf{e}_1 - 2\mathbf{e}_2$ but leaves $\mathbf{e}_2$ unchanged, find the standard

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

matrix A of the transformation. Then, find $T\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right)$.

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} \quad A = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) = 2\begin{bmatrix} 1 \\ -2 \end{bmatrix} + 3\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \end{bmatrix} + \begin{bmatrix} 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

7. (4 points) Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation where $T(\mathbf{u}) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and

$T(\mathbf{v}) = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$. Use the fact that T is linear to find $T(\mathbf{u} + 3\mathbf{v})$.

$$T(\mathbf{u}) + 3T(\mathbf{v}) \rightarrow \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 3\begin{bmatrix} 1 \\ -3 \end{bmatrix} \rightarrow \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 3 \\ -9 \end{bmatrix} \rightarrow \begin{bmatrix} 5 \\ -8 \end{bmatrix}$$

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8. For this problem, consider the matrix $A = [a_1 \ a_2 \ a_3] = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 2 & 1 \\ 3 & 1 & 1 \end{bmatrix}$, which can be row

reduced into the echelon form $\begin{bmatrix} 1 & 3 & 1 \\ 0 & -4 & -1 \\ 0 & 0 & 0 \end{bmatrix}$. (You do not need to verify this.)

- a) (5 points) Do the columns of A span $\mathbb{R}^3$? Justify your answer using words.

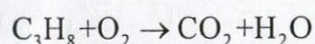
No due to $\mathbb{R}_3$ being all zeros

- b) (5 points) Is $v = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$ in $\text{Span}\{a_1, a_2, a_3\}$? If so, show how v can be written as a linear combination of a_1 , a_2 , and a_3 .

v is a multiple of a_1 meaning it is in $\text{span}\{a_1, a_2, a_3\}$

$$2 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + 0 \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

9. (8 points) When propane gas burns, the propane (C_3H_8) combines with oxygen (O_2) to form carbon dioxide (CO_2) and water (H_2O) according to an equation of the form



Set up an appropriate vector equation which could be used to balance this chemical equation. There is no need to actually solve the vector equation. Be sure that you clearly indicate what the entries in the vectors represent.

$$\begin{bmatrix} C \\ H \\ O \end{bmatrix} \times \begin{bmatrix} 3 \\ 8 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ -2 \end{bmatrix} + x_4 \begin{bmatrix} 0 \\ -2 \\ -1 \end{bmatrix}$$

$$\hookrightarrow x_1 \begin{bmatrix} 3 \\ 8 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ -2 \end{bmatrix} + x_4 \begin{bmatrix} 0 \\ -2 \\ -1 \end{bmatrix} = 0$$

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10. Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$, and $\mathbf{v}_4 = \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}$.

a) (6 points) Is $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ linearly dependent or linearly independent?

Use the method of your choice to justify your answer.

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \xrightarrow{R_2 + R_1 \rightarrow R_2} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix} \xrightarrow{R_3 - R_1 \rightarrow R_3} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 3 \\ 0 & -1 & 1 \end{bmatrix} \xrightarrow{R_2 - R_1 \rightarrow R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

Independent, no free variables $\left(\begin{smallmatrix} - \\ 1 \end{smallmatrix} \right)$

b) (6 points) Is $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ linearly dependent or linearly independent?

Justify your answer in words. Do not perform row reductions.

Dependent, too many entries for the number of rows

c) (6 points) Is $\mathbf{v}_4$ in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$? Justify your answer in words.

$$\begin{bmatrix} 1 & 2 & 0 & -2 \\ -1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

No, there is not a pivot in every row AND column.

11. (4 points) The following *augmented* matrix in echelon form represents a system of linear equations. If the system is consistent, state the general solution in parametric vector form. Otherwise, state that the system is inconsistent and clearly indicate why you came to that conclusion.

$$\left[\begin{array}{ccc|c} 1 & 2 & -7 & 5 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

Inconsistent, there's a pivot in the augmented column

$$\rightarrow 0 + 0 + 0 \neq 1$$

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$$\begin{aligned} x_1 + 2x_3 &= 5 & x_1 &= 5 - 2x_3 \\ x_2 - 3x_3 &= 5 & x_2 &= 5 + 3x_3 \\ x_3 &= x_3 \end{aligned}$$

12. a) (14 points) Solve the matrix equation $Ax = b$ using augmented matrices, where

$$A = \begin{bmatrix} 1 & 2 & -4 \\ 0 & 1 & -3 \\ 2 & 1 & 1 \end{bmatrix} \text{ and } b = \begin{bmatrix} 3 \\ -1 \\ 9 \end{bmatrix}.$$

Show all your work! State your solution in parametric vector form.

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 2 & -4 & 3 \\ 0 & 1 & -3 & -1 \\ 2 & 1 & 1 & 9 \end{array} \right] & \xrightarrow{R_2 - 2R_1 \rightarrow R_2} \left[\begin{array}{ccc|c} 1 & 2 & -4 & 3 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \end{array} \right] \\ & \xrightarrow{-\frac{1}{3}R_2} \left[\begin{array}{ccc|c} 1 & 2 & -4 & 3 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \end{array} \right] \xrightarrow{R_3 - R_2 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & 2 & -4 & 3 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right] \\ & \xrightarrow{R_1 - 2R_2 \rightarrow R_1} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 5 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right] \end{aligned}$$

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- b) (4 points) Without performing any additional row operations, state the solution set for the homogeneous equation $Ax = 0$ where A is the matrix given in part (a) above.

$$\begin{bmatrix} 1 & 2 & -4 \\ 0 & 1 & -3 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} 5 - 2x_3 \\ 5 + 3x_3 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad -3$$

13. (4 points) Construct a nonzero 2×2 matrix A such that the vector $\begin{bmatrix} 3 \\ -1 \end{bmatrix}$ is

a solution of $Ax = 0$.

$$\begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} 3x_1 - x_3 &= 0 \\ 3x_2 - x_4 &= 0 \end{aligned}$$

$$x_1 = \frac{1}{3}x_3$$

$$x_2 = \frac{1}{3}x_4$$